

# **Smart Study Assistant – Feature List**

## **1. Intelligent Study Lamp (Voice & Gesture Controlled)**

Components: ESP32 / Arduino, Microphone/Sound Sensor, Webcam, LDR Sensor, Servo Motor, LED/Buzzer

AI/ML: OpenCV-based hand detection

Functionality:

- Voice command activates system
- Detects hand position and adjusts lamp direction
- Maintains proper lighting using LDR input
- Auto timeout and return to idle mode

## **2. Attention Detection System**

Components: Microcontroller (ESP32/Arduino), Webcam, LED/Buzzer

AI/ML: OpenCV-based face detection and gaze estimation

Functionality:

- Tracks user face orientation
- Detects focus vs distraction
- Triggers alert on distraction

## **3. Temperature & Humidity Monitoring**

Components: Microcontroller, DHT11/DHT22 Sensor, LED/Buzzer, Optional Display

Functionality:

- Monitors environmental conditions

- Alerts when temperature or humidity exceeds safe range

#### **4. Pomodoro-Based Study Timer**

Components: Microcontroller, Push Button, LED/Buzzer

Functionality:

- Tracks study sessions (25 min study + 5 min break)
- Provides alerts for session transitions
- Optional logging of study activity

#### **5. LLM-Based Voice Assistant (Basic)**

Components: Microcontroller + PC/Server Interface, Microphone, Speaker

AI/ML: Integration with LLM (e.g., via API) + Speech-to-Text / Text-to-Speech

Functionality:

- Accepts voice commands/questions
- Provides simple responses (study help, timer control, system status)
- Can control system features via voice

#### **6. OCR to PDF Generator (Handwritten Notes)**

Components: Webcam, PC Connection, Microcontroller (trigger/control)

AI/ML: OCR model (e.g., Tesseract OCR or deep learning-based handwritten text recognition)

Functionality:

- Captures handwritten text via camera
- Converts image to digital text
- Generates and saves PDF when connected to PC